

Introduction

This project analyzes **energy consumption patterns** in **City of Toronto–owned and operated buildings**.

We use the *Annual Energy Consumption* dataset to study how **electricity and natural gas usage** vary by:

- building type
- building size
- time of year (month)

The project focuses on understanding **month-to-month trends** using **data science models** to estimate how much energy a building is expected to consume in a given month.

Hypothesis / Goal

Project Goal

The goal of this project is to understand how **month, building type, and building size** affect energy consumption, and to use this information to **predict monthly electricity and natural gas usage** for city-owned buildings. This goal will help us predict future energy consumption for city budgeting and long term energy-usage planning. This goal will also help us identify major energy consumers, inefficiencies and unusual energy usage patterns localized to city owned/operated buildings. These issues if addressed will align with one of Toronto's long term goals of building a sustainable future.

Key Tentative Hypotheses

- **H₁ (Natural Gas):**
Monthly natural gas consumption is significantly higher during winter months than during summer months, even after accounting for building type and size.
 - **H₂ (Electricity):**
Monthly electricity consumption varies by building type and month, but shows weaker seasonal variation compared to natural gas consumption.
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Dataset and Preprocessing

Dataset

We use the **Annual Energy Consumption – City Owned/Operated Buildings** dataset provided by the City of Toronto.

The dataset includes:

- monthly electricity and natural gas usage
- energy cost
- meter type
- building information such as property type and gross floor area

Each energy record represents **one building over one billing period (month)**.

Preprocessing Steps

- Extract month and year from billing dates
- Performing data deduplication if required
- Handle different meter types by modeling **electricity and gas separately**
- Identify and exclude invalid floor area values (e.g., 1 ft²) from area-based analysis
- Optionally normalize usage by billing period length (usage per day)

Ethics Challenges

The dataset contains **only aggregated energy usage data** for publicly owned buildings and does **not include any personal or sensitive information**.

However:

- The data represents only **city-owned properties**, so results may not generalize to private buildings.
 - Some building attributes (such as floor area) contain placeholder or missing values, which must be handled carefully to avoid misleading conclusions.
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Algorithm Selection

Regression Models (Main Focus)

We use **linear regression** to build two predictive models:

1. Electricity Consumption Model

- Predicts monthly electricity usage (kWh)
- Features: building type, floor area, month

2. Natural Gas Consumption Model

- Predicts monthly gas usage (m³)
- Features: building type, floor area, month

Regression is appropriate because it allows us to:

- quantify relationships between features and energy use
- estimate expected consumption for a given building and time of year

Clustering Model (Supporting Analysis)

We apply **K-means clustering** to group buildings based on their energy consumption patterns.

Clustering helps identify:

- high gas– buildings with strong usage
- high electricity– buildings with strong usage
- low overall energy consumers

This provides **structural insight** that complements the regression models.

Visualization

The following visualizations are used to support analysis and model interpretation:

1. **Seasonal Trend Line Chart**

- Average monthly electricity and gas consumption across all buildings
- Shows seasonal patterns and motivates regression features

2. **Monthly Distribution (Boxplots)**

- Distribution of energy usage by month
- Highlights variability and outliers

3. **Regression Relationship Scatter Plot**

- Floor area vs energy consumption with regression line
- Shows relationship between size and usage

4. **Clustering Profile Plot**

- Average monthly energy usage for each cluster
- Visualizes distinct energy consumption behaviors

Conclusion

In this project, we analyze monthly/seasonal energy consumption patterns in city-owned buildings using **regression and clustering techniques**.

Regression models allow us to **predict monthly electricity and natural gas usage** based on building characteristics and time of year, while optional clustering reveals **groups of buildings with similar energy behaviors**.

Together, these approaches provide meaningful insights into how energy is consumed across the city and lay the foundation for future work on **energy efficiency and planning**.

Questions

Thank you for your attention, are there any questions?